Model Program Book



SEMESTER INTERNSHIP

Designed & Developed by

ANDHRA PRADESH

STATE COUNCIL OF HIGHER EDUCATION

PROGRAM BOOK FOR

SEMESTER INTERNSHIP

Name of Student : BUDIGI VAMSHI

Name of College : SAI RAJESWARI INSTITUTE OF TECHNOLOGY

Registration Number : 228R1A0514

Period of Internship : **From :** 24th Nov 2025 **To :** 31st Mar 2026

Name & Address of The Intern Organization : Codeworks Edutech Services Pvt. Ltd., Intern Stump Platform

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A Statutory Body of Government of Andhra Pradesh)

PROGRAM BOOK FOR SEMESTER INTERNSHIP

An Internship Report on: CYBER SECURITY

Submitted in partial fulfillment of the requirement for the degree of

B.Tech (Computer Science & Engineering)

Under the Faculty Guideship of

Dr. Prabakaran Narayan , M.Tech, Ph.D

Professor , Department of CSE

Submitted by:

Budigi Vamshi : 228R1A0514

Department of Computer Science & Engineering



SAI RAJESWARI INSTITUTE OF TECHNOLOGY

AN ISO 9001:2015 CERTIFIED INSTITUTION

(Approved by AICTE, New Delhi and Affiliated to J.N.T.U.A. Ananthapuramu) Lingapuram (V), Proddatur, Kadapa(Dist.)-516360

2022 - 2026

Internship Period: 24th November 2025 to 31st March 2026

Organization: Codeworks EduTech Services Pvt. Ltd.

Certificate ID: CWSRITIN0024

codeworks

AN ISO 9001:2015 CERTIFIED COMPANY

NATIONAL INTERNSHIP PORTAL

Certificate Of Internship

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A Statutory Body of Government of the Andhra Pradesh)

This is to certify that

Ms./Mr. BUDIGI VAMSHI , Regd. No.: 228R1A0514

under **SAI RAJESWARI INSTITUTION OF TECHNOLOGY** of **JNTU Anantapur** has successfully completed a **Offline Longterm-Term Internship** on **CYBER SECURITY** from **24th Nov 2025**to **31st March 2026**. Organized by **Codeworks EduTech Services Pvt. Ltd.**in collaboration with **Andhra Pradesh State Council of Higher Education**.

DATE OF ISSUE: 31st March 2026

CERTIFICATE ID : CWSRITIN0024

HEAD OF THE TRAININGS

KORRAPATI RAVI THARUN
CEO & MD

Certifications & Recognizations

#startupindia | MSME | ISO 9001:2015 | Ministry of Commerce and Industry | अखिल भारतीय तकनीकी शिक्षा परिषद् All India Council for Technical Education

campus@codeworkstech.com | www.codeworkstech.com | /codeworksedutech

Student's Declaration

I, Budigi Vamshi, a student of B.Tech (Computer Science & Engineering), Reg. No. 228R1A0514, of the Department of Computer Science & Engineering , Sai Rajeswari Institute of Technology, do hereby declare that I have completed the mandatory internship from 24th November 2025 to 31st March 2026 in Codeworks EduTech Services Pvt. Ltd. (Intern Stump) under the Faculty Guideship of (Dr. Prabakaran Narayan), Department of Computer Science & Engineering, Sai Rajeswari Institution of Technology.

Signature & Date: _____

Official Certification

This is to certify that Budigi Vamshi, Reg. No. 228R1A0514, has completed his Internship in Codeworks EduTech Services Pvt. Ltd. on CYBER SECURITY under the supervision of the authorized mentor, as a part of partial fulfillment of the requirement for the Degree of B.Tech (CSE) in the Department of Computer Science & Engineering, Sai Rajeswari Institute of Technology.

This is accepted for evaluation.

Signatory with Date and Seal: _____

Endorsements

Faculty Guide: _____

Head of the Department: _____

Principal: _____

Certificate from Intern Organization

This is to certify that Budigi Vamshi, Reg. No. 228R1A0514, of Sai Rajeswari Institute of Technology, underwent internship in Codeworks EduTech Services Pvt. Ltd. (Intern Stump) from 24th November 2025 to 31st March 2026.

The overall performance of the intern during his/her internship is found to be: Satisfactory

Authorized Signatory with Date and Seal: _____

Acknowledgements

I express my sincere gratitude to Codeworks EduTech Services Pvt. Ltd. for providing me the opportunity to undertake this Cyber Security internship. I am thankful to the Head of Trainings, the CEO & MD Mr. Korrapati Ravi Tharun, and all the team members at Intern Stump for their constant guidance and support during the internship period from 24th November 2025 to 31st March 2026. I also extend my heartfelt thanks to my Faculty Guide and the Principal of Sai Rajeswari Institute of Technology, JNTU Anantapur, for their encouragement throughout this program.

Chapter 1: Executive Summary

This report documents the Semester Long-Term Internship completed by Budigi Vamshi (Reg. No. 228R1A0514) as a Cyber Security Intern at Codeworks EduTech Services Pvt. Ltd. (Intern Stump), organized in collaboration with the Andhra Pradesh State Council of Higher Education (APSCHE). The internship was conducted in Offline Long-Term mode from 24th November 2025 to 31st March 2026, spanning approximately 18 weeks.

Learning Objectives and Outcomes

1. Understanding and applying network security, vulnerability assessment, and penetration testing techniques using industry-standard tools.
2. Developing proficiency in security assessment tools such as Kali Linux, Nmap, Metasploit, Burp Suite, and Wireshark.
3. Acquiring hands-on experience in ethical hacking, exploit development, and security incident documentation.
4. Strengthening communication skills through professional security report writing and threat briefings.
5. Gaining practical knowledge of real-world security frameworks including MITRE ATT&CK, OWASP Top 10, and PTES methodology.
6. Understanding organizational security policies, risk management, and professional cybersecurity workplace practices.

Organization Overview

Codeworks EduTech Services Pvt. Ltd. is an ISO 9001:2015 certified EdTech company recognized under Startup India and the Ministry of MSME. It operates the 'Intern Stump' platform that connects students with industry-relevant internship programs. The organization is accredited by AICTE and the All India Council for Technical Education.

Internship Summary

During this internship, the intern was exposed to various aspects of cybersecurity including network security, vulnerability assessment, web application security, penetration testing, and security reporting. The internship involved working on simulated lab environments to identify and exploit vulnerabilities, creating security assessment reports, and presenting findings. The intern also participated in team discussions and practical case study analysis, which contributed to both technical and professional development.

Chapter 2: Overview of the Organization

A. Introduction of the Organization

Codeworks EduTech Services Pvt. Ltd. is a leading EdTech services provider operating under the brand 'Intern Stump.' It is an ISO 9001:2015 certified company recognized by the Government of India under Startup India and MSME initiatives. The company collaborates with Andhra Pradesh State Council of Higher Education (APSCHE) to offer structured, industry-relevant internship programs to engineering and degree students across Andhra Pradesh and Telangana.

B. Vision, Mission, and Values

Vision: To bridge the gap between academic learning and industry expectations by providing quality internship experiences.

Mission: To empower students with practical skills, industry exposure, and professional competencies that enable them to contribute meaningfully in their careers.

Values: Excellence, Integrity, Innovation, Student-Centric approach, and Continuous Improvement.

C. Organizational Structure

The organization is led by the CEO & MD (Mr. Korrapati Ravi Tharun) and the Head of Trainings. It comprises departments for Training & Internship Coordination, Industry Relations, Student Support, Technology, and Administration.

D. Roles and Responsibilities Relevant to Intern

As a Cyber Security Intern, the primary responsibilities included conducting vulnerability assessments, performing ethical hacking exercises, analyzing security logs, preparing security reports, participating in weekly reviews, and contributing to case study analysis projects as part of the 8C appraisal framework.

Chapter 3: Internship Activity Logs & Weekly Reports

This chapter documents the week-by-week activity logs and detailed weekly reports of the internship from 24th November 2025 to 31st March 2026.

ACTIVITY LOG FOR THE FIRST WEEK

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person In Charge Signature
Day - 1 24-11-25 (Mon)	Orientation session: Introduction to Codeworks EduTech, team members, and internship workflow. Overview of Cyber Security role expectations and responsibilities.	Understood organizational structure, communication protocols, and internship objectives in cybersecurity context.	
Day - 2 25-11-25 (Tue)	Attended a workshop on professional communication in cybersecurity: email etiquette for security alerts, incident report writing, and presenting threat findings to stakeholders.	Learned to frame security insights clearly in written and verbal formats suitable for both technical and non-technical audiences.	
Day - 3 26-11-25 (Wed)	Practice session: Drafted a sample security incident report and presented it to the mentor for feedback. Focused on clarity, conciseness, and completeness of security information.	Improved ability to structure security reports; received constructive feedback on simplifying complex technical findings.	
Day - 4 27-11-25 (Thu)	Participated in a team discussion on an assigned security scenario. Practiced active listening, asking clarifying questions, and summarizing key threat points.	Developed active listening and oral communication skills; learned to contribute constructively in security team meetings.	
Day - 5 28-11-25 (Fri)	Reviewed sample security dashboards and practiced explaining the visualizations using simple, clear language suitable for management and business stakeholders.	Gained skills in communicating security metrics verbally and adapting technical language to the audience.	
Day - 6 29-11-25 (Sat)	Weekly review meeting with supervisor. Presented Week 1 learnings and received feedback on communication style and security report drafting.	Understood areas for improvement in communication; built confidence in presenting security findings to supervisors.	

WEEKLY REPORT -- WEEK 1 (From: 24-11-2025 To: 29-11-2025)

Objective of the Activity Done (1C: Communication):

To develop professional communication skills essential for a Cyber Security analyst — including written reporting of vulnerabilities, verbal briefings on security threats, and effective participation in security team discussions.

Detailed Report:

Week 1 of the internship was dedicated to the 1C competency: Communication. The week began with an orientation session introducing the intern to Codeworks EduTech, the Intern Stump platform, team members, and the expectations of the Cyber Security internship program.

The intern attended a structured workshop on professional communication in cybersecurity, covering how to write security incident reports, vulnerability disclosures, and executive-level threat briefings. A key focus was on tailoring technical security content for non-technical stakeholders.

A hands-on exercise required the intern to draft a sample security incident summary report. The supervisor reviewed the draft and provided feedback on clarity, professional tone, and completeness. The intern revised and resubmitted an improved version.

ACTIVITY LOG FOR THE SECOND WEEK

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person In Charge Signature
Day - 1 01-12-25 (Mon)	Introduction to Kali Linux: installation and configuration of Kali Linux in VirtualBox VM. Configured network settings (NAT + Host-only adapter) for isolated lab environment.	Learned to set up and configure a standard cybersecurity lab environment using virtualization.	
Day - 2 02-12-25 (Tue)	Installed and configured Metasploitable 2 as a vulnerable target VM. Set up network connectivity between Kali Linux and target machine. Verified connectivity using ping and Nmap.	Gained hands-on experience in configuring a safe, isolated penetration testing lab environment.	
Day - 3 03-12-25 (Wed)	Configured Burp Suite as a web proxy interceptor. Set up browser certificate and proxy settings. Created first intercepted HTTP request to verify configuration.	Understood Burp Suite configuration, proxy interception, and basic web application security testing setup.	
Day - 4 04-12-25 (Thu)	Configured Wireshark for network traffic capture. Set up capture filters for specific protocols. Captured and analyzed first network traffic session.	Acquired practical skills in configuring network packet capture tools for traffic analysis.	
Day - 5 05-12-25 (Fri)	Configured a shared Google Drive workspace for the team. Organized folder structure for security reports, scripts, and research. Set up naming conventions for security documentation.	Developed skills in configuring collaborative cloud workspaces and maintaining organized security documentation.	
Day - 6 06-12-25 (Sat)	Weekly review: demonstrated all configured environments to supervisor. Resolved configuration issues identified during review. Documented the complete setup process for reference.	Reinforced troubleshooting skills; produced a configuration guide document for future reference.	

WEEKLY REPORT -- WEEK 2 (From: 01-12-2025 To: 06-12-2025)

Objective of the Activity Done (2C: Configuration):

To configure all required tools, environments, and workspaces for the Cyber Security internship, including Kali Linux, network analysis tools, vulnerability scanners, and collaborative platforms.

Detailed Report:

Week 2 focused on the 2C competency: Configuration. The primary goal was to establish a fully functional and well-organized technical environment for cybersecurity work throughout the internship.

The intern set up the Kali Linux environment — the industry-standard OS for cybersecurity professionals. Essential tools including Nmap, Wireshark, Metasploit Framework, Burp Suite Community Edition, and John the Ripper were installed and configured. Each tool was tested with a sample exercise to verify correct setup.

A virtualization environment was configured using VirtualBox, with separate VMs for Kali Linux (attack machine) and an intentionally vulnerable target machine (Metasploitable 2). This provided a safe, isolated lab environment for security testing exercises.

ACTIVITY LOG FOR THE THIRD WEEK

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person InCharge Signature
Day - 1 08-12-25 (Mon)	Introduction to chronological security logging: maintaining daily activity logs, timestamped security event records, and incident journals using structured templates aligned with SOC practices.	Learned the importance of chronological security documentation and practised using structured log templates.	
Day - 2 09-12-25 (Tue)	Worked with security log datasets: loaded and explored Linux system logs, Windows Event Logs, and web server access logs. Practiced parsing timestamps and sorting events chronologically.	Gained skills in handling and analyzing log data using command-line tools (grep, awk, tail, journalctl).	
Day - 3 10-12-25 (Wed)	Incident timeline reconstruction: analyzed a simulated security incident log dataset. Built a chronological attack timeline identifying initial access, privilege escalation, and lateral movement stages.	Developed ability to reconstruct security incident timelines and map events to MITRE ATT&CK techniques.	
Day - 4 11-12-25 (Thu)	Practiced maintaining a detailed security work journal: recorded all tasks, tools used, findings, and blockers for the day in a structured format reviewed by the supervisor.	Understood how chronological security records support incident documentation, accountability, and continuous improvement.	
Day - 5 12-12-25 (Fri)	Analyzed a historical security event dataset to track attack progression over time. Calculated MTTD and MTTR metrics, and identified recurring attack patterns in the log timeline.	Applied chronological analysis techniques to derive actionable security insights from historical event data.	
Day - 6 13-12-25 (Sat)	Presented a chronological security event analysis report to the team covering attack timelines and log findings from the week. Received feedback on documentation quality.	Improved report presentation skills; understood best practices for chronological security event storytelling.	

WEEKLY REPORT -- WEEK 3 (From: 08-12-2025 To: 13-12-2025)

Objective of the Activity Done (3C: Chronical):

To develop skills in chronological security logging, incident timeline reconstruction, and structured documentation — ensuring accurate tracking of security events, activities, and outcomes over time.

Detailed Report:

Week 3 was aligned with the 3C competency: Chronical — focusing on the ability to record, track, and analyze security events and activities in a time-ordered, systematic manner.

The week began with training on structured security documentation practices. The intern was introduced to security log templates requiring timestamped entries, event descriptions, severity levels, and response actions. This practice is fundamental in real-world security operations centers (SOC).

A major technical focus was security log analysis. The intern worked with system logs, application logs, and network logs using tools like Splunk (free trial) and Linux command-line utilities (grep, awk, tail). Log parsing exercises helped build the ability to reconstruct incident timelines from raw log data.

ACTIVITY LOG FOR THE FOURTH WEEK

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person InCharge Signature
Day - 1 15-12-25 (Mon)	Introduction to completeness in cybersecurity: studied comprehensive attack surface enumeration, threat coverage models, and the impact of incomplete security assessments on organizational risk.	Understood the types and consequences of incomplete security coverage and the importance of thorough assessment methodologies.	
Day - 2 16-12-25 (Tue)	Complete port and service enumeration using Nmap: conducted full TCP/UDP scans on the target VM. Documented all open ports, services, and versions discovered.	Gained practical experience in comprehensive network enumeration to ensure complete attack surface coverage.	
Day - 3 17-12-25 (Wed)	Conducted a complete web application security scan using Nikto and OWASP ZAP on the target's web services. Documented all identified vulnerabilities and misconfigurations.	Developed ability to conduct thorough web application security assessments and produce structured vulnerability reports.	
Day - 4 18-12-25 (Thu)	Practiced end-to-end completeness checks on security assessment findings: verified all CVSS categories were assessed, identified scan gaps, and documented findings comprehensively.	Learned to apply systematic completeness checks across all dimensions of a security assessment (network, web, system, application).	
Day - 5 19-12-25 (Fri)	Created a vulnerability completeness dashboard in Excel showing severity distribution, coverage rates by asset, and remediation tracking to ensure no vulnerability was overlooked.	Built skills in visualizing vulnerability completeness metrics to support security governance and risk management.	
Day - 6 20-12-25 (Sat)	Week review: Presented the complete vulnerability assessment report and coverage dashboard to the supervisor. Discussed implications of any coverage gaps for organizational security posture.	Understood how assessment completeness directly impacts security outcomes and organizational risk decisions.	

WEEKLY REPORT -- WEEK 4 (From: 15-12-2025 To: 20-12-2025)

Objective of the Activity Done (4C: Completeness):

To develop a thorough understanding of completeness in cybersecurity — ensuring comprehensive vulnerability assessments, complete security reports, and thorough coverage of all attack surfaces.

Detailed Report:

Week 4 addressed the 4C competency: Completeness — a foundational quality dimension in cybersecurity that directly influences the effectiveness of security assessments and the reliability of defensive recommendations.

The week began with a conceptual overview of completeness in cybersecurity. The intern studied the importance of thorough attack surface enumeration, ensuring no vulnerabilities are missed during security assessments, and the consequences of incomplete security testing (e.g., undetected attack vectors).

The primary hands-on work involved conducting a comprehensive vulnerability assessment of the Metasploitable 2 target machine. The intern used Nmap for complete port and service enumeration, Nikto for

ACTIVITY LOG FOR THE FIFTH WEEK

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person InCharge Signature
Day - 1 22-12-25 (Mon)	Introduction to security cognizance: studying the MITRE ATT&CK framework, threat intelligence concepts, and the importance of situational awareness in cybersecurity operations.	Gained awareness of the importance of understanding the threat landscape, attacker TTPs, and organizational risk context.	
Day - 2 23-12-25 (Tue)	Studied CVE databases (NIST NVD), threat intelligence feeds, and CISA security advisories. Mapped current high-severity vulnerabilities to the simulated lab environment.	Developed the ability to interpret and work with threat intelligence sources and CVE databases effectively.	
Day - 3 24-12-25 (Wed)	OSINT reconnaissance exercise: applied tools (theHarvester, Maltego Community, Shodan) to gather open-source intelligence on a fictional target organization for security assessment context.	Practised OSINT techniques to build comprehensive cognizance of target attack surface using publicly available information.	
Day - 4 25-12-25 (Thu)	Public Holiday (Christmas). Self-study: reviewed online resources on threat intelligence frameworks, MITRE ATT&CK, and Cyber Kill Chain methodology.	Expanded conceptual understanding of threat modeling and cybersecurity cognizance frameworks.	
Day - 5 26-12-25 (Fri)	Conducted a threat profiling exercise for the simulated environment: documented threat actors, attack vectors, MITRE ATT&CK technique mappings, and risk severity ratings.	Learned systematic threat profiling to rapidly build cognizance of the security risk context for any target environment.	
Day - 6 27-12-25 (Sat)	Presented a threat cognizance report to the team: summarised threat landscape findings, identified top risks, and proposed prioritized defensive actions.	Developed the ability to synthesise threat intelligence into concise, actionable cognizance reports.	

WEEKLY REPORT -- WEEK 5 (From: 22-12-2025 To: 27-12-2025)

Objective of the Activity Done (5C: Cognizance):

To build deep cognizance of the cybersecurity threat landscape — understanding attack techniques, threat intelligence, current vulnerabilities, and the organizational security context.

Detailed Report:

Week 5 corresponded to the 5C competency: Cognizance — the ability to develop thorough, contextual awareness of the cybersecurity landscape before and during security activities. Cognizance in cybersecurity encompasses threat intelligence, knowledge of attack techniques, and situational awareness.

The week began with an introduction to threat intelligence frameworks. The intern studied the MITRE ATT&CK framework in depth — learning the tactics, techniques, and procedures (TTPs) used by threat actors. This provided a structured mental model for understanding how attacks unfold.

A key activity was studying the organization's threat landscape. The intern reviewed current CVE (Common Vulnerabilities and Exposures) databases, threat intelligence feeds, and security advisories from CISA and NIST. This built awareness of real-world threats relevant to the EdTech sector.

ACTIVITY LOG FOR THE SIXTH WEEK

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person InCharge Signature
Day - 1 29-12-25 (Mon)	Introduction to security assessment conception: translating organizational security requirements into a penetration testing framework. Studied scoping techniques and rules of engagement.	Learned to formulate clear security assessment objectives, define scope boundaries, and design rules of engagement.	
Day - 2 30-12-25 (Tue)	Worked on a case study: given a fictional EdTech company scenario experiencing data breaches, conceptualised a complete penetration testing plan — defined scope, methodology, and expected deliverables.	Developed skills in security assessment planning: scope definition, methodology selection, and deliverable specification.	
Day - 3 31-12-25 (Wed)	Designed a conceptual security architecture: identified security controls needed at network, application, and endpoint layers. Mapped controls to NIST Cybersecurity Framework categories.	Understood how to design conceptual security architectures aligned with threat models and compliance frameworks.	
Day - 4 01-01-26 (Thu)	Public Holiday (New Year). Self-study: reviewed case studies of security assessment conception from industry (web app pentesting, network pentesting, red team engagements).	Gained insight into real-world security assessment conception processes and how business context shapes testing design.	
Day - 5 02-01-26 (Fri)	Developed a penetration testing blueprint: outlined all PTES methodology phases (Pre-engagement, Intelligence Gathering, Threat Modeling, Exploitation, Post-Exploitation, Reporting).	Acquired skills in designing end-to-end security assessment blueprints aligning testing workflows with client objectives.	
Day - 6 03-01-26 (Sat)	Presented the security assessment concept and blueprint to the supervisor. Received feedback on feasibility, scope, and alignment with organizational risk management objectives.	Improved ability to communicate security assessment concepts to stakeholders and refine plans based on feedback.	

WEEKLY REPORT -- WEEK 6 (From: 29-12-2025 To: 03-01-2026)

Objective of the Activity Done (6C: Conception):

To develop the ability to conceive structured security solutions — translating organizational security requirements into clear penetration testing plans, conceptual security architectures, and end-to-end assessment blueprints.

Detailed Report:

Week 6 focused on the 6C competency: Conception — the cognitive and planning process of designing a security assessment or defensive solution before any technical execution begins. This is a critical skill distinguishing a competent cybersecurity professional from one who relies purely on tool execution.

The week opened with training on security assessment scoping techniques. The intern was taught to define the scope of a penetration test — identifying in-scope assets, defining testing boundaries, documenting rules of engagement, and obtaining conceptual authorization (simulated).

A case study exercise involved a scenario of a fictional EdTech company experiencing data breaches. The intern developed a complete penetration testing concept document: defining objectives, specifying in-scope systems, identifying testing methodologies (black-box, grey-box), and outlining deliverables.

Security architecture conception was introduced. The intern designed a conceptual security architecture for the fictional company — identifying security controls needed at network, application, and endpoint layers based on a threat model.

An assessment blueprint was produced that mapped the complete penetration testing workflow from scoping and reconnaissance through exploitation, post-exploitation, and final reporting. This blueprint followed industry-standard PTES (Penetration Testing Execution Standard) methodology.

The week concluded with a presentation of the conceived security assessment plan. The supervisor's feedback highlighted areas for scope refinement and provided valuable lessons in aligning security testing plans with business objectives and risk tolerance.

ACTIVITY LOG FOR THE SEVENTH WEEK

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person InCharge Signature
Day - 1 05-01-26 (Mon)	Deep-dive into vulnerability comprehension: studied how to interpret CVE advisories, understand CVSS scores, and contextualize vulnerabilities within organizational risk frameworks.	Developed the ability to critically interpret vulnerability advisories, CVSS metrics, and their business risk implications.	
Day - 2 06-01-26 (Tue)	Exploitation comprehension: used Metasploit Framework to execute exploitation modules on Metasploitable 2. Studied each exploit's mechanism, payload options, and post-exploitation capabilities.	Gained hands-on experience in exploitation and deep comprehension of how vulnerabilities are weaponized by attackers.	
Day - 3 07-01-26 (Wed)	Practiced understanding OWASP Top 10 vulnerabilities on DVWA: exploited SQL Injection, XSS, and Command Injection with focus on understanding root causes and remediation strategies.	Improved ability to comprehend web application vulnerability classes, their exploitation logic, and defensive countermeasures.	
Day - 4 08-01-26 (Thu)	Studied network-level attack comprehension: analyzed ARP spoofing and Man-in-the-Middle attack mechanics using Wireshark captures. Understood traffic interception at the packet level.	Strengthened network security comprehension skills: understanding attack mechanisms at the protocol and packet level.	
Day - 5 09-01-26 (Fri)	Comprehension exercise: read and analyzed a published security research paper on a real-world attack campaign. Summarized methodology, exploitation chain, indicators of compromise, and defensive measures.	Enhanced ability to comprehend and critically evaluate external security research and apply learnings to own assessments.	
Day - 6 10-01-26 (Sat)	Presented a comprehensive security analysis summary to the team: demonstrated understanding of all topics covered during weeks 1-7 and their interconnections in a complete attack-defense framework.	Consolidated comprehension of the full cybersecurity workflow; demonstrated ability to connect concepts across attack and defense.	

WEEKLY REPORT -- WEEK 7 (From: 05-01-2026 To: 10-01-2026)

Objective of the Activity Done (7C: Comprehension):

To develop strong security comprehension — the ability to deeply understand attack techniques, vulnerability details, exploitation mechanisms, and their real-world security implications.

Detailed Report:

Week 7 was dedicated to the 7C competency: Comprehension — the ability to deeply understand cybersecurity concepts, technical attack mechanisms, and their real-world defensive implications, going beyond surface-level tool usage.

The week began with a focus on vulnerability comprehension. The intern revisited discovered vulnerabilities with a deeper emphasis on understanding root causes — understanding why a buffer overflow occurs, how SQL injection exploits trust boundaries, and what conditions enable privilege escalation.

ACTIVITY LOG FOR THE EIGHTH WEEK

Day & Date	Brief Description of Daily Activity	Learning Outcome	Person In Charge Signature
Day - 1 12-01-26 (Mon)	Introduction to penetration testing case study methodology: studied PTES standard phases, reviewed real-world pentest report examples, and defined scope and objectives for the week's case study.	Understood structured penetration testing methodologies and how to apply them systematically to case study environments.	
Day - 2 13-01-26 (Tue)	Case Study Day 1: Conducted reconnaissance and enumeration on the simulated EdTech company environment. Performed full Nmap scans, web application fingerprinting, and OSINT gathering.	Applied reconnaissance skills to a real-world case study environment; identified key services and preliminary attack vectors.	
Day - 3 14-01-26 (Wed)	Case Study Day 2: Performed vulnerability analysis on discovered services. Used Nikto, OpenVAS, and manual testing to identify and verify exploitable vulnerabilities across web and network layers.	Practised end-to-end vulnerability analysis on a simulated environment, prioritizing findings by severity and exploitability.	
Day - 4 15-01-26 (Thu)	Case Study Day 3: Executed controlled exploitation of identified vulnerabilities — exploited web RCE, weak database credentials, and outdated SSH service. Documented all exploitation steps with evidence.	Applied exploitation techniques in a controlled environment; demonstrated impact of discovered vulnerabilities with documented proof-of-concept.	
Day - 5 16-01-26 (Fri)	Case Study Day 4: Produced a professional penetration testing report — Executive Summary, detailed findings with CVSS scores, evidence screenshots, remediation recommendations, and risk summary.	Developed skills in translating technical security findings into professional, actionable penetration testing reports.	
Day - 6 17-01-26 (Sat)	Case Study Presentation: Presented the complete penetration testing case study to the supervisor and team. Answered technical questions and demonstrated all 8C competencies through the integrated deliverable.	Successfully completed a full end-to-end penetration testing case study; demonstrated integration of all 8C competencies.	

WEEKLY REPORT -- WEEK 8 (From: 12-01-2026 To: 17-01-2026)

Objective of the Activity Done (8C: Case Study Analysis):

To apply all previously developed competencies (Communication, Configuration, Chronical, Completeness, Cognizance, Conception, and Comprehension) in a structured, real-world cybersecurity case study — producing actionable security findings and professional-grade deliverables.

Detailed Report:

Week 8 represented the culmination of the 8C appraisal framework through the 8C: Case Study Analysis competency. This week integrated all skills developed across the previous seven weeks into a comprehensive, end-to-end cybersecurity case study.

The case study was based on a simulated EdTech company environment provided by Codeworks EduTech, comprising a web application, database server, and internal network segment. The objective was to conduct a full penetration test, identify vulnerabilities, and provide actionable remediation recommendations.

Following the PTES (Penetration Testing Execution Standard) methodology, the intern progressed through all phases: Pre-engagement (defining scope and rules of engagement), Intelligence Gathering (OSINT and active reconnaissance), Threat Modeling (identifying high-value targets), Exploitation (controlled exploitation of discovered vulnerabilities), Post-Exploitation (privilege escalation and lateral movement assessment), and Reporting.

The exploitation phase yielded several findings: an unauthenticated Remote Code Execution vulnerability in a web service, weak credential configurations on database services, and an outdated SSH version susceptible to known exploits. All findings were exploited in the controlled lab environment with documented evidence.

A professional penetration testing report was produced covering Executive Summary, Scope and Methodology, Detailed Findings (with CVSS scores, evidence screenshots, and remediation guidance), and a Risk Summary Dashboard. The report followed industry-standard penetration testing report structure.

The case study presentation was well-received by the supervisor, who commended the thoroughness of the assessment, the quality of the report, and the practical, business-relevant nature of the remediation recommendations. This marked the successful completion of the 8C appraisal framework.

Chapter 5: Outcomes Description

Work Environment Experienced

The work environment at Codeworks EduTech Services Pvt. Ltd. was highly professional and collaborative. The organization maintained clear job roles, structured daily schedules, and an open culture for communication. Facilities were well maintained and conducive to learning. Team interactions were professional and supportive, with regular check-ins and progress reviews conducted by the Head of Trainings.

Real-Time Technical Skills Acquired

During the internship, hands-on skills were acquired in Kali Linux, Nmap, Metasploit Framework, Burp Suite, Wireshark, OpenVAS, and OWASP ZAP for security assessment. Real-time simulated lab environments were used for practice, which helped in developing a strong foundation in cybersecurity assessment and penetration testing workflows.

Managerial Skills Acquired

The internship helped in developing time management, goal-setting, and planning skills. Working in a team environment improved leadership ability and decision-making. Weekly reporting practices improved performance tracking and self-assessment.

Communication Skills Improvement

Regular presentations of security findings to supervisors and team members significantly improved both oral and written communication abilities. Structured security report formats enhanced the ability to articulate vulnerabilities and remediation recommendations clearly, concisely, and confidently.

Group Discussion and Team Participation

Active participation in group discussions, threat modeling sessions, and team security project meetings enhanced collaboration skills. Contributing to team activities as both a team member and occasionally as a discussion leader helped develop interpersonal competencies.

Technological Developments Observed

During the internship, emerging trends in cybersecurity such as AI-powered threat detection, cloud security posture management (CSPM), zero-trust architecture implementation, and automated vulnerability management platforms were observed. These technologies are rapidly transforming the role of cybersecurity professionals in modern organizations.

Student Self Evaluation of the Long-Term Internship

Student Name: Budigi Vamshi
Registration No: 228R1A0514
Term of Internship (From): 24-11-2025
Term of Internship (To): 31-03-2026
Date of Evaluation: 31-03-2026
Organization: Codeworks EduTech Services Pvt. Ltd. (Intern Stump)
Internship Title: CYBER SECURITY

Please rate your performance in the following areas (1 = Lowest, 5 = Highest):

Sl.No	Evaluation Area	1	2	3	4	5
1	Oral Communication	1	2	3	4	5
2	Written Communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction Ability with Community	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Self-Confidence	1	2	3	4	5
7	Ability to Learn	1	2	3	4	5
8	Work Plan and Organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
10	Creativity	1	2	3	4	5
11	Quality of Work Done	1	2	3	4	5
12	Time Management	1	2	3	4	5
13	Understanding the Community	1	2	3	4	5
14	Achievement of Desired Outcomes	1	2	3	4	5
15	**OVERALL PERFORMANCE**	1	2	3	4	5

Date: _____ Signature of the Student: _____

Evaluation by the Supervisor of the Intern Organization

Student Name: Budigi Vamshi
Registration No: 228R1A0514
Term of Internship (From): 24-11-2025
Term of Internship (To): 31-03-2026
Date of Evaluation: 31-03-2026
Organization: Codeworks EduTech Services Pvt. Ltd. (Intern Stump)

Please rate the student's performance in the following areas (1 = Lowest, 5 = Highest):

Sl.No	Evaluation Area	1	2	3	4	5
1	Oral Communication	1	2	3	4	5
2	Written Communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction Ability with Community	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Self-Confidence	1	2	3	4	5
7	Ability to Learn	1	2	3	4	5
8	Work Plan and Organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
10	Creativity	1	2	3	4	5
11	Quality of Work Done	1	2	3	4	5
12	Time Management	1	2	3	4	5
13	Understanding the Community	1	2	3	4	5
14	Achievement of Desired Outcomes	1	2	3	4	5
15	**OVERALL PERFORMANCE**	1	2	3	4	5

Date: _____ Signature of the Supervisor: _____

8C Appraisal Record — Codeworks EduTech (Intern Stump)

This is to certify that Budigi Vamshi has successfully completed the CYBER SECURITY internship at Intern Stump from 24-11-2025 to 31-03-2026. The 8C Appraisal Record is outlined below:

Code	Competency	Rating (out of 5)
1C	Communication	/5
2C	Configuration	/5
3C	Chronical	/5
4C	Completeness	/5
5C	Cognizance	/5
6C	Conception	/5
7C	Comprehension	/5
8C	Case Study Analysis	/5

Evaluation & Marks Statement

Internal Assessment Statement

Student Name: Budigi Vamshi
Registration No: 228R1A0514
Programme of Study: B.Tech (Information Technology)
Year of Study: 3rd / 4th Year
College: Sai Rajeswari Institution of Technology
University: JNTU Anantapur

Sl.No	Evaluation Criterion	Maximum Marks	Marks Awarded
1.	Activity Log	10	
2.	Internship Evaluation	30	
3.	Oral Presentation	10	
	GRAND TOTAL	50	

Date: _____ Signature of the Faculty Guide: _____

External Assessment Statement

Sl.No	Evaluation Criterion	Maximum Marks	Marks Awarded
1.	Internship Evaluation	80	
2.	Supervisor Grading (Intern Org)	20	
3.	Viva-Voce	50	
	TOTAL	150	
	GRAND TOTAL (Internal 50 + External 150)	200	

Signature of the Faculty Guide: _____

Signature of the Internal Expert: _____

Signature of the External Expert: _____

Signature of the Principal with Seal: _____

ANDHRA PRADESH
STATE COUNCIL OF HIGHER EDUCATION

(A Statuory Body of the Government of Andhra Pradesh)

2nd, 3rd, 4th and 5th floors, Neeladri Towers, Sri Ram Nagar, 6th Battalion Road
Atmakur (V)Mangalagiri (M), Guntur, Andhra Pradesh, Pin - 522 503
www.apsche.ap.gov.in